Master's Comprehensive Examination

Part I Theory (90 minutes) December 6, 1997

Instructions: Answer all four questions. Open book and open notes.

1. A box is known to contain either 3 red and 5 black balls or 5 red and 3 black balls. Three balls are drawn from the box without replacement and if less than 3 red balls are obtained, the decision will be made that the box contains 3 red and 5 black balls. Calculate $P(I)$ and $P(II)$, the probabilities of type I and type II errors. (Answer need not be in most simple form.)

2. (a) In estimating the mean of a normal variable by means of a confidence interval, how large a sample will be needed if the length of a 95% confidence interval is to be less than $\sigma/10$ if σ is assumed known?

(b) Let X be normally distributed with mean 0 and variance 1. That is, its density is $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$. Find the following conditional probability: $P[X < 1 | X > 0]$.

3. Let $X_1, X_2, \ldots, X_m$ be a random sample of size m from $N(\theta, \sigma^2)$.

Let $Y_1, Y_2, \ldots, Y_n$ be a random sample of size n from $N(\theta + \gamma, \sigma^2)$ with X's and Y's independent of each other. Find a 95% confidence interval for γ, assuming θ, γ, and σ^2 are all unknown.

4. Given a random sample of size n from a normal distribution $N(0, \sigma^2)$ find the maximum likelihood estimator of σ^2. That is $X_1, \ldots, X_n \sim N(0, \sigma^2)$ so that $f_{X_i}(x_i; \sigma^2) = \frac{1}{\sqrt{2\pi}(\sigma^2)^{1/2}} e^{-\frac{x_i^2}{2\sigma^2}}$.

Master's Comprehensive Examination

Part II | Applied (90 minutes) | December 6, 1997

Instructions: Answer all four questions below. Open book and open notes.

1. A pharmaceutical company is interested in comparing two drugs with respect to the reduction in blood pressure. Each patient uses each drug for one month with a one month washout period between each drug use. The response variable is the reduction in systolic blood pressure from the beginning to the end of each month of drug use. The data from the experiment are shown below.

Patient	Drug A	Drug B
1	-20	-15
2	-25	-5
3	-10	-15
4	-15	0
5	-10	-15
6	-20	-20
7	-15	-20
8	-25	-15
9	-30	-15
10	-10	-10

Assuming the data above are normally distributed

a) Set up the appropriate null and alternative hypothesis.

b) Using the appropriate test statistic, test the hypothesis formulated in (a). Let $\alpha = 0.01$.

c) If the true difference between the drugs is 10, with drug A showing a greater reduction than drug B, what is the power of the test used in (b).

d) Construct a 95% confidence interval for the mean difference between the drugs.

2. The following experiments assume that an analysis of variance is to be performed. For each experiment list the sources of variation, the degrees of freedom, and the expected

mean square. Be sure to indicate which sources of variation are fixed, which are random, and which are mixed.

a) An experimenter wants to compare six temperatures with respect to the reaction rate of a chemical experiment. On each of four days the experimenter makes seven determinations of reaction rate, one for each of the seven temperatures.

b) An experimenter wants to compare six temperatures with respect to the reaction rate of a chemical experiment. On each of four days the experimenter makes 14 determinations of reaction rate, two for each of the seven temperatures.

c) An experimenter wants to compare six temperatures with respect to the reaction rate of a chemical experiment. Three laboratory technicians were asked to run the six temperatures on each of four randomly selected days.

3. Two experiments are conducted to determine the skin sensitivity of a new cream.

a) In the first experiment, 200 subjects are used, 100 subjects placing cream A on their left hand and the other 100 subjects placing cream B on their left hand. After two days each subject is evaluated for a skin reaction. The results of their experiment are given below.

		Cream A	Cream B
Reaction	Yes	30	50
	No	70	50

b) In the second experiment, 100 subjects are used, with each subject placing cream A on one hand and cream B on the other hand. The assignment of cream to hand was done randomly. The results of this experiment are given below showing the effect of both products on each subject.

Reaction-Cream A Reaction-Cream B	No Reaction-Cream A Reaction-Cream B
30	20

Reaction-Cream A No Reaction-Cream B	No Reaction-Cream A No Reaction-Cream B
40	10

For each experiment:

a) Set up the appropriate null and alternative hypothesis.

b) Using the appropriate test statistic, test the hypothesis formulated in (a). Let $\alpha = 0.05$.

c) Construct a 95% confidence interval for the difference between the creams with respect to the proportion of skin reactions.

4. An experiment was conducted to develop a dose response relationship between insecticide concentration and kill rate. Kill rate was determined by subjecting 20 insects to the insecticide and counting how many were dead within one hour. The data collected were

X (concentration)	Y (proportion dead)
.001	0.00
.01	0.10
.1	0.30
1	0.40
10	0.60
100	0.70
1000	0.90

a) Write down the appropriate model that describes the relationship between X and Y.

b) Estimate the parameters of the model.

c) Develop a statistic which demonstrates how well the model characterizes the relationship between x and y.

d) Construct a hypothesis which tests the significance of the relationship between X and Y.

e) Test the hypothesis given in (d).

Master's Comprehensive Examination

Part I

Theory (90 minutes)

March 8, 1997

Instructions: Answer all four questions. Open book and open notes.

1. Let X be a random variable with density $f_X(x|\lambda) = \frac{1}{\lambda}e^{-\frac{x}{\lambda}}$, $x > 0$, $\lambda > 0$.

(a) Find the cumulative distribution function of X at $x = 1$. That is, find $P(X \leq 1)$.

(b) Find the coefficient of variation. That is, find

$$r = \frac{\text{standard deviation of } X}{\text{mean of } X}.$$

2. (a) For the model of question 1, find the best test of size .05 of the hypothesis $H_0 : \lambda = 1$ vs $H_1 : \lambda = 2$.

(b) Is this the same as the best test for $H_1 : \lambda = 1$ vs $H_a : \lambda = 7$? Give a reason for your answer.

3. (a) In a box of 100 items, I sample 5 without replacement. 90 of the items are good but 10 are defective. If I find 3 or more defectives I will discard the box. Find the probability of discarding the box.

(b) What is the answer if I sample with replacement?

4. Let $X_1, \ldots, X_n$ be a sample of size n from a normal distribution with mean 0 and variance τ^2.

(a) Find the MLE, maximum likelihood estimator, of τ^2.

(b) Is it unbiased? Prove your answer.

Master's Comprehensive Examination

Part II | Applied (90 minutes) | March 8, 1997

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. An experiment was performed using eight tomato plants. One half of the flowers of each plant was pollinated by the experimenter using a new pollinating technique. The seed yield from each half of the plant is tabulated below.

Plant no.	Seed yield from experimental pollinating technique	Seed yield from standard pollinating technique
1	10	8
2	12	14
3	8	6
4	12	10
5	14	10
6	8	8
7	10	12
8	12	10

(a) Determine if the experimental technique gives a higher yield in seed than the standard technique (let $\alpha = .05$). Be sure that both H_0 and H_1 are shown.

(b) How can the expeiimenter make a Type I error? What are the consequences of his doing so?

(c) How can the experimenter make a Type II error? What are the consequences of his doing so?

(d) Give a 90% confidence interval for the difference between the two techniques.

2. A random sample of 300 families is obtained and the number of children each family has is recorded. The results of this study are summarized in the following table.

Number of Children	0	1	2	3	4	5	6	7	8
Number of Families	6	24	42	59	62	44	41	14	8

Use a goodness-of-fit test to determine if these data follow a Poisson distribution.

3. A simple linear regression model

$$Y = \alpha + \beta X + \epsilon$$

was fit to a set of data. The following results were found.

$b = 4$

$\bar{X} = 4$

$\bar{Y} = 20$

Additionally part of the analysis of variance table is summarized below.

Source	df	SS	MS	F
Total	10	800		
Regression				
Residual		288		

(a) Find the estimated regression line.

(b) If it is assumed that ϵ is distributed as $N(0, \sigma^2)$, then estimate σ^2.

(c) Test the hypothesis that $\beta = 0$. Let the probability of a Type I error $= .10$.

(d) Find a 95% confidence interval for the mean value of Y when $X = 8$.

(e) Find the coefficient of determination. What does this mean?

(f) Find an estimate of the correlation coefficient, ρ.

4. For each of the following situations, assume that an Analysis of Variance is to be performed. List the sources of variation, the degrees of freedom and the expected mean square. Be sure to indicate which sources of variation are fixed and which are random.

(a) An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. Each of 6 state experiment stations has agreed to grow the 3 varieties. Each station will homogenize a randomly selected tomato of each variety, determine the percent acid content, and submit this one measurement (for each variety) to the experimenter.

(b) An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. From a single experiment station he obtains a bushel of each variety. For each variety he then selects 6 tomatoes at random from the bushel and records the percent acid content of each of the selected tomatoes.

(c) An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. In a greenhouse, 4 benches were used. Placed on each bench was a single plant from each variety. From each plant 2 fruit were selected and the percent acid content of each fruit was recorded.

(d) An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. He finds 6 home gardeners who are growing variety A, 6 who are growing variety B, and 6 who are growing variety C. Each gardener will supply the experimenter with 2 tomatoes and the experimenter will then record a measure of percent acid content for each tomato.

5. In reading Pap smears, pathologists have difficulty in agreeing on whether a certain minor atypical cellular appearance is abnormal or not. Two pathologists classified the same 52 slides independently, grading them zero, indicating normal, to 4, indicating invasive carcinoma. The results are as follows.

		Pathologist A					
		0	1	2	3	4	
Pathologist B	0	21	3	1	0	0	25
	1	1	4	2	1	0	8
	2	0	2	6	4	0	12
	3	0	0	2	2	0	4
	4	0	0	0	0	3	3
		22	9	11	7	3	52

Is there evidence that one pathologist tends to read more pathology than the other? (Test the hypothesis that neither sees more disease than the other. Compute the P-value.)